Muhammed Roshan M K

Software Engineer | Python | FastAPI | Backend Development
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PROFESSIONAL SUMMARY

Backend Software Engineer with 2.5+ years building production-grade, cloud-native Python systems for regulated healthcare products. Designs async REST APIs and microservices with FastAPI, SQLAlchemy, and Pydantic, backed by PostgreSQL, Redis, and Kafka. Recently delivered patient-facing mobile backends with AI RAG chatbot orchestration, self-hosted Chatwoot on GCP, Firebase single-device login, and scheduled push notifications. Owns delivery end to end, from architecture and data modelling to JWT/RBAC security, Docker, CI/CD, and GCP deployment.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java
Backend: FastAPI, REST API Design, API Versioning, Microservices, Async/Await, Pydantic, Dependency Injection, Middleware, Background Tasks, OpenAPI/Swagger
Databases: PostgreSQL, MySQL, SQLite, SQLAlchemy ORM, Alembic, Schema Design, Indexing, Query Optimization
Caching & Messaging: Redis, Apache Kafka, Event-Driven Architecture
AI & Mobile Backend: RAG, LangChain, Chroma (Vector Search), Embeddings, Whisper (Speech-to-Text), LLM Orchestration (Gemini), Chatwoot, Firebase Cloud Messaging, Firebase Auth
Computer Vision: YOLO (Ultralytics), PyTorch, ONNX / ONNX Runtime, OpenCV, Object Tracking (ByteTrack, SORT), ffmpeg, WebGPU
Cloud & DevOps: GCP (Compute Engine, Cloud Storage, Artifact Registry), AWS (basic), Docker, Docker Compose, Bitbucket Pipelines, CI/CD, Poetry, Git, Render, Vercel
Practices: JWT Authentication, RBAC, Repository & Service Layer Patterns, Structured Logging, Code Reviews, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer | Gadgeon Smart Systems Pvt. Ltd., Kochi, Kerala **Jan 2024 – Present**

PregPath App – Patient Mobile App with AI RAG Chatbot

- Built the backend for the patient-facing mobile companion to the PregPath clinical platform, extending care directly to expectant mothers.
- Provisioned a GCP Compute Engine VM hosting self-hosted Chatwoot with Redis and PostgreSQL, keeping full control of patient conversation data.
- Developed a chat orchestrator middleware on GCP routing messages between Chatwoot and the AI RAG chatbot for context-aware patient answers.
- Designed weekly AI-driven check-in flows that collect patient status updates and surface health signals to the clinical team.
- Implemented single-device login with Firebase device tokens and FCM push notifications for scheduled weekly check-in reminders.

PregPath AI – AI-Driven Maternal Healthcare Platform

- Designed async, versioned FastAPI REST APIs for real-time maternal health monitoring and risk assessment, with strict Pydantic validation.
- Architected modular microservices (core API, rule engine, dashboard API) with clear service boundaries and shared contracts.
- Built Kafka producers and consumers to decouple long-running risk assessments from the request path; added Redis caching with TTL and invalidation.
- Evolved the PostgreSQL schema with SQLAlchemy and Alembic; tuned indexes and resolved N+1 queries on high-volume clinical tables.
- Packaged a shared data-access layer as a versioned Poetry module used by three services, published to Artifact Registry via Bitbucket Pipelines.
- Secured protected health data with JWT authentication and RBAC enforced through reusable FastAPI dependencies and middleware.

Medical AI – AI-Driven General Healthcare Platform

- Built FastAPI services for patient registration, diagnosis monitoring, treatment tracking, and recovery, with paginated, filterable endpoints.
- Developed an automated report service that renders clinical PDFs as background tasks and uploads them to Google Cloud Storage.
- Fixed a critical reporting defect by migrating from FPDF to ReportLab with custom paragraph-breaking, eliminating pagination failures.

- Containerized services with Docker Compose, added structured logging and centralized exception handling, and led root-cause analysis with QA and DevOps.

PERSONAL PROJECTS

VisionTrack – Real-Time Object Detection & Tracking **Live Demo | GitHub**

Python, YOLO, PyTorch, ONNX Runtime, OpenCV, FastAPI, ffmpeg, WebGPU, Docker

- Built real-time object detection and multi-object tracking for images, short videos, and live webcam, running one YOLO26n model in PyTorch locally, ONNX Runtime on the server, and WebGPU in the browser (~20 FPS).
- Exported the model to ONNX and reimplemented pre/post-processing in NumPy to match Ultralytics exactly, removing PyTorch from the server (~306 MB peak RAM on a 512 MB free host); wrote a SORT-style tracker and an ffmpeg H.264 video job pipeline.

DocChat – RAG Document Assistant **Live Demo | GitHub**

Python, FastAPI, LangChain, Chroma, Gemini, Docker

- Built a RAG chatbot that answers questions from uploaded PDFs, articles, and notes with inline source and page citations, streamed over SSE.
- Implemented chunking, local ONNX embeddings, and Chroma vector search; cut peak memory from ~2 GB to ~360 MB for a 512 MB free host, with per-session isolation, rate limiting, and SSRF protection.

MeetScribe – AI Meeting Transcription Assistant **Live Demo | GitHub**

Python, FastAPI, Whisper, ffmpeg, Groq, Gemini, Docker

- Built a pipeline that turns meeting recordings into timestamped transcripts, summaries, decisions, and action items, exportable as Markdown, TXT, and SRT.
- Processed audio with ffmpeg into overlapping chunks, transcribed with Whisper large-v3, and generated Pydantic-validated summaries with map-reduce in background jobs.

EDUCATION

B.Tech in Computer Science | LBS College of Engineering, Kasaragod, Kerala **Jul 2019 – Jun 2023**